

Ch 11: Electricity

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. What is electric current? Write its SI unit. How is current related to charge and time? (PYQ Pattern)

Q2. Define one ampere of current. A charge of 10 C flows through a wire in 5 s. Find the current. (PYQ Pattern)

Q3. What is electric potential difference? Write its SI unit. Define one volt. (PYQ Pattern)

Q4. Name the instrument used to measure potential difference. How is it connected in a circuit? (PYQ Pattern)

Q5. Name the instrument used to measure current. How is it connected in a circuit? (PYQ Pattern)

Q6. State Ohm's law. Write the mathematical relation between V, I and R. (PYQ Pattern)

Q7. What is resistance? Write its SI unit. Define one ohm. (PYQ Pattern)

Q8. Draw the symbols of: (i) electric cell (ii) battery (iii) resistor (iv) key/switch. (PYQ Pattern)

Q9. Draw the symbols of: (i) ammeter (ii) voltmeter (iii) rheostat (iv) connecting wire. (PYQ Pattern)

Q10. What is meant by a closed circuit and an open circuit? Give one difference. (PYQ Pattern)

Q11. What is a circuit diagram? Why are standard symbols used in circuit diagrams? (PYQ Pattern)

Q12. What is a rheostat? State its function in an electric circuit. (PYQ Pattern)

Q13. Why is an ammeter connected in series in a circuit? (PYQ Pattern)

Q14. Why is a voltmeter connected in parallel across a component? (PYQ Pattern)

Q15. A potential difference of 12 V is applied across a resistor of 6 ohm. Calculate the current. (PYQ Pattern)

Q16. A current of 0.5 A flows through a conductor when 3 V is applied across it. Find its resistance. (PYQ Pattern)

Q17. What happens to current when potential difference is doubled, resistance remaining constant? Give reason. (PYQ Pattern)

Q18. What happens to resistance of a wire when its length is doubled, area remaining constant? (PYQ Pattern)

Q19. What happens to resistance of a wire when its area of cross-section is doubled, length remaining constant? (PYQ Pattern)

Q20. Write the factors on which the resistance of a conductor depends. (PYQ Pattern)

Q21. What is resistivity? Write its SI unit. Does resistivity depend on length and area of the conductor? (PYQ Pattern)

Q22. Why are copper and aluminium generally used for electrical transmission wires? (PYQ Pattern)

Q23. Why is nichrome used in electric heating devices? Give two reasons. (PYQ Pattern)

Q24. What is the equivalent resistance when two resistors R_1 and R_2 are connected in series? (PYQ Pattern)

Q25. What is the equivalent resistance when two resistors R_1 and R_2 are connected in parallel? (PYQ Pattern)

Q26. In a series circuit, how is current distributed through different resistors? (PYQ Pattern)

Q27. In a parallel circuit, how is potential difference distributed across different resistors? (PYQ Pattern)

Q28. Two resistors of 2 ohm and 3 ohm are connected in series. Find the equivalent resistance. (PYQ Pattern)

Q29. Two resistors of 2 ohm and 3 ohm are connected in parallel. Find the equivalent resistance. (PYQ Pattern)

Q30. Why is parallel arrangement preferred in household electrical circuits? Give two reasons. (PYQ Pattern)

Q31. What is heating effect of electric current? Name one device based on this effect. (PYQ Pattern)

Q32. State Joule's law of heating. Write the formula for heat produced in a resistor. (PYQ Pattern)

Q33. What is electric power? Write its SI unit. Define one watt. (PYQ Pattern)

Q34. Write three formulae for electric power in terms of V , I and R . (PYQ Pattern)

Q35. Define one kilowatt hour. What is its relation with joule? (PYQ Pattern)

Q36. An electric bulb is rated 60 W, 220 V. What does this rating mean? (PYQ Pattern)

Q37. Why does the filament of an electric bulb glow while connecting wires do not glow? (PYQ Pattern)

Q38. Why are coils of electric toasters and electric irons made of alloys rather than pure metals? (PYQ Pattern)

Q39. A fuse wire has low melting point. Why is this property useful? (PYQ Pattern)

Q40. Write the difference between resistance and resistivity. Give one point each. (PYQ Pattern)

SECTION B - 3 MARKS

Q1. Derive the relation $I = Q/t$. Explain the meaning and SI unit of each symbol. (PYQ Pattern)

Q2. Explain potential difference between two points. How is work done related to charge and potential difference? (PYQ Pattern)

Q3. State Ohm's law. Describe an experiment to verify Ohm's law using ammeter and voltmeter. (PYQ Pattern)

Q4. Draw a circuit diagram to verify Ohm's law and mention how the graph between V and I appears. (PYQ Pattern)

Q5. A current of 2 A flows through a resistor of 10 ohm for 5 minutes. Calculate the charge flowing through it. (PYQ Pattern)

Q6. A charge of 600 C flows through a circuit in 10 minutes. Calculate the current flowing in the circuit. (PYQ Pattern)

Q7. A 4 ohm resistor is connected to a 12 V battery. Calculate: (i) current (ii) charge flowing in 2 minutes. (PYQ Pattern)

Q8. Explain why resistance of a conductor depends on length, area of cross-section, nature of material and temperature. (PYQ)

Pattern)

Q9. A wire of resistance R is stretched to double its length. What happens to its resistance if volume remains constant? Explain.

(PYQ Pattern)

Q10. Define resistivity. Write the relation $R = \rho l/A$ and explain each term. (PYQ Pattern)

Q11. Compare conductors, resistors and insulators on the basis of resistance and use. Give one example each. (PYQ Pattern)

Q12. Why are alloys preferred in heating elements? Explain with reference to resistivity, melting point and oxidation. (PYQ Pattern)

Q13. Derive the formula for equivalent resistance of three resistors connected in series. (PYQ Pattern)

Q14. Three resistors of 2 ohm, 3 ohm and 5 ohm are connected in series with a 10 V battery. Find total resistance and current. (PYQ Pattern)

Q15. In a series combination of resistors, show that the total resistance is greater than the greatest individual resistance. (PYQ Pattern)

Q16. Derive the formula for equivalent resistance of three resistors connected in parallel. (PYQ Pattern)

Q17. Three resistors of 2 ohm, 3 ohm and 6 ohm are connected in parallel. Find the equivalent resistance. (PYQ Pattern)

Q18. In a parallel combination of resistors, show that the equivalent resistance is less than the least individual resistance. (PYQ Pattern)

Q19. Compare series and parallel combinations of resistors in terms of current, voltage and equivalent resistance. (PYQ Pattern)

Q20. Why is series arrangement not used for domestic appliances? Give three reasons. (PYQ Pattern)

Q21. A 6 ohm and a 3 ohm resistor are connected in parallel to a 6 V battery. Calculate total current drawn from the battery. (PYQ Pattern)

Q22. A 5 ohm resistor and a 10 ohm resistor are connected in series to a 30 V battery. Find current and potential difference across each resistor. (PYQ Pattern)

Q23. State Joule's law of heating. Explain how heat depends on current, resistance and time. (PYQ Pattern)

Q24. Derive $H = I^2Rt$ using the relation between work, charge, current and potential difference. (PYQ Pattern)

Q25. An electric heater of resistance 20 ohm draws a current of 5 A for 2 minutes. Calculate heat produced. (PYQ Pattern)

Q26. A bulb of resistance 440 ohm is connected to a 220 V supply. Calculate current and power consumed. (PYQ Pattern)

Q27. Explain the working of an electric fuse. Why should a fuse wire be connected in series with the appliance? (PYQ Pattern)

Q28. What is electric power? Derive $P = VI$, $P = I^2R$ and $P = V^2/R$. (PYQ Pattern)

Q29. A device rated 100 W is used for 5 hours daily. Calculate the energy consumed in kWh in 10 days. (PYQ Pattern)

Q30. An electric lamp is marked 100 W, 220 V. Calculate: (i) current drawn (ii) resistance of filament when glowing. (PYQ Pattern)

Q31. Compare 1 watt, 1 kilowatt and 1 kilowatt hour. State which one is a unit of energy. (PYQ Pattern)

Q32. Why does a high-power appliance draw more current than a low-power appliance at the same voltage? Explain using formula. (PYQ Pattern)

Q33. Draw a labelled circuit containing a battery, key, ammeter, resistor and voltmeter. Show correct connection of ammeter and voltmeter. (PYQ Pattern)

Q34. What happens to the heat produced in a resistor if: (i) current is doubled (ii) resistance is doubled (iii) time is doubled? (PYQ Pattern)

Q35. Answer the following: (i) Why is tungsten used in bulb filaments? (ii) Why is copper used in wires? (iii) Why is ebonite an insulator? (PYQ Pattern)

SECTION C - 5 MARKS

Q1. Explain electric current and potential difference in detail. Define ampere and volt. Derive the relation $V = W/Q$ and solve: 300 J of work is done to move 60 C charge between two points. Find the potential difference. (PYQ Pattern)

Q2. State Ohm's law. Describe the complete experimental verification with circuit diagram, observation table and V-I graph. Mention the condition under which Ohm's law is obeyed. (PYQ Pattern)

Q3. A conductor obeys Ohm's law. Explain how current changes with potential difference. Draw the V-I graph and calculate resistance if current is 0.4 A at 8 V. (PYQ Pattern)

Q4. Explain resistance of a conductor. Derive and discuss R proportional to l , R proportional to $1/A$ and $R = \rho l/A$. State SI unit of resistivity. (PYQ Pattern)

Q5. A wire of length 2 m and area of cross-section $1 \times 10^{-6} \text{ m}^2$ has resistance 4 ohm. Calculate its resistivity. Also explain whether resistivity changes if length is doubled. (PYQ Pattern)

Q6. Compare copper, nichrome and rubber as materials used in electrical circuits. Include resistance/resistivity, role and one practical use of each. (PYQ Pattern)

Q7. Derive the expression for equivalent resistance of resistors connected in series. Using the result, find total resistance and current when 2 ohm, 4 ohm and 6 ohm are connected in series to a 24 V battery. (PYQ Pattern)

Q8. Derive the expression for equivalent resistance of resistors connected in parallel. Using the result, find equivalent resistance when 3 ohm, 6 ohm and 9 ohm are connected in parallel. (PYQ Pattern)

Q9. Compare series and parallel combinations in detail: current, voltage, equivalent resistance, failure of one component and domestic use. (PYQ Pattern)

Q10. A 3 ohm and 6 ohm resistor are connected in parallel. This combination is connected in series with a 2 ohm resistor and a 12 V battery. Find total resistance, total current and current through each parallel branch. (PYQ Pattern)

Q11. Two resistors of 4 ohm and 8 ohm are first connected in series and then in parallel across the same battery of 24 V. Calculate current in each arrangement and compare the equivalent resistances. (PYQ Pattern)

Q12. Three resistors of 5 ohm, 10 ohm and 30 ohm are connected in parallel to a 15 V supply. Calculate equivalent resistance, total current and current through each resistor. (PYQ Pattern)

Q13. Explain heating effect of electric current. Derive Joule's law of heating $H = I^2 R t$ from electrical work done. State three applications of heating effect. (PYQ Pattern)

Q14. An electric iron of resistance 44 ohm is connected to a 220 V supply for 10 minutes. Calculate: (i) current (ii) power (iii) heat produced. (PYQ Pattern)

Q15. A heater produces 72000 J of heat in 2 minutes when connected to a 120 V source. Find power, current and resistance of the heater. (PYQ Pattern)

Q16. Explain electric power in detail. Derive $P = VI$, $P = I^2R$ and $P = V^2/R$. Give the SI unit of power and commercial unit of electrical energy. (PYQ Pattern)

Q17. An electric bulb is marked 220 V, 100 W. Calculate its resistance and current. If it is used for 5 hours daily for 30 days, calculate energy consumed in kWh. (PYQ Pattern)

Q18. A household uses five 60 W bulbs for 6 hours daily, one 1000 W heater for 2 hours daily and one 100 W fan for 10 hours daily. Calculate total electrical energy consumed in one day in kWh. (PYQ Pattern)

Q19. Explain why domestic appliances are connected in parallel and not in series. Include voltage, independent switching, equivalent resistance and appliance safety. (PYQ Pattern)

Q20. A fuse of 5 A is connected in a circuit. Can appliances of total power 1500 W be safely used on a 220 V supply? Calculate and justify. (PYQ Pattern)

Q21. Draw a neat circuit diagram to study the dependence of current on potential difference. Label battery, key, rheostat, ammeter, resistor and voltmeter. Explain the role of each component. (PYQ Pattern)

Q22. Case Study: A student connects three resistors of 2 ohm, 4 ohm and 4 ohm in a circuit with a 12 V battery. Answer: (i) equivalent resistance in series (ii) current in series (iii) equivalent resistance in parallel (iv) total current in parallel (v) which combination draws more current and why. (PYQ Pattern)

Q23. Case Study: An electric kettle is rated 2000 W, 220 V.

Answer: (i) meaning of rating (ii) current drawn (iii) resistance (iv) energy used in 30 minutes (v) why thick wires are used for high-power appliances. (PYQ Pattern)

Q24. Answer the following numericals in one solution: (i) Find current when 24 C charge flows in 6 s. (ii) Find resistance when 10 V produces 2 A. (iii) Find heat when 3 A flows through 5 ohm for 20 s. (iv) Find power when 220 V draws 0.5 A. (PYQ Pattern)

Q25. Write a comprehensive note on electricity covering: electric current, potential difference, resistance, Ohm's law, resistivity, series combination, parallel combination, heating effect and electric power with formulae. (PYQ Pattern)

Q26. A circuit contains a 6 ohm resistor in series with a parallel combination of 3 ohm and 6 ohm. The circuit is connected to a 16 V battery. Calculate equivalent resistance, total current, voltage across series resistor and voltage across parallel branch. (PYQ Pattern)